



# Vegf signaling between Müller glia and vascular endothelial cells is regulated by immune cells and stimulates retina regeneration

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In the zebrafish retina, Müller glia (MG) can regenerate retinal neurons lost to injury or disease. Even though zebrafish MG share structure and function with those of mammals, only in zebrafish do MG function as retinal stem cells. Previous studies suggest dying neurons, microglia/macrophage, and T cells contribute to MG's regenerative response [White et al., *Proc. Natl. Acad. Sci. U.S.A.* 114, E3719 (2017); Hui et al., *Dev. Cell* 43, 659 (2017)]. Although MG end-feet abut vascular endothelial (VE) cells to form the blood–retina barrier, a role for VE cells in retina regeneration has not been explored. Here, we report that MG-derived Vegfaa and Pgfα engage Flt1 and Kdr1 receptors on VE cells to regulate MG gene expression, Notch signaling, proliferation, and neuronal regeneration. Remarkably, *vegfaa* and *pgfα* expression is regulated by microglia/macrophages, while Notch signaling in MG is regulated by a Vegf-*dll4* signaling system in VE cells. Thus, our studies link microglia/macrophage, MG, and VE cells in a multicomponent signaling pathway that controls MG reprogramming and proliferation.

stem cell | Notch | reprogramming | neurodegeneration | Pgf

In response to injury, zebrafish Müller glia (MG) undergo a partial reprogramming event that imparts them with retinal stem cell characteristics (1, 2). Reprogrammed MG divide asymmetrically to generate a multipotent progenitor capable of regenerating all major retinal neuron types (3–5). Previous studies have identified several signal-transduction cascades and gene expression programs that regulate MG reprogramming and proliferation (6–20); many of which are regulated by factors derived from MG, nerve, and immune cells (7–9, 11, 13, 20–23). In addition to these interactions, MG end-feet abut vascular endothelial (VE) cells lining blood vessels to form the blood–retina barrier (24). This barrier responds to changes in neuronal activity and neurodegeneration that can lead to altered vascular flow, permeability, and sprouting (25–27). Vegf signaling is the predominant pathway regulating VE cell growth, and increased Vegf expression is associated with increased retinal angiogenesis and disease. We recently reported that neurodegeneration in the zebrafish retina regulates gene expression in VE cells (14). Thus, VE cells might impact MG's regenerative response. Here, we investigate this possibility using needle poke, NMDA, and UV light treatments to stimulate retinal neuron and photoreceptor-specific degeneration in adult zebrafish.

## Results

**Vegf Signaling Stimulates MG Proliferation in the Injured Retina.** We first investigated if Vegf signaling regulates MG proliferation in the injured retina by incubating fish in the pan Vegf receptor (Vegfr) inhibitor, pazopanib (28), and assaying PCNA immunofluorescence on retinal sections at 2 d post injury (dpi) (Fig. 1A). Quantification of PCNA+ cells in the INL revealed a dose-dependent decrease in MG proliferation following Vegfr blockade (Fig. 1A). As previously reported (5), we confirm *1016 tuba1a:GFP*+ MG are the predominant cell type to proliferate in the injured retina's inner nuclear layer (INL) (arrows in *SI Appendix, Fig. S1 A, Left*). Other nonretinal cell types like microglia/macrophages (4C4+) and MG-derived rod progenitors [round nuclei in the outer nuclear layer (ONL)] are sometimes observed to proliferate in the injured retina at a low level (arrowheads in *SI Appendix, Fig. S1 A*), while 4C4+ microglia/macrophages and *flt1:EGFP*+ VE cells have not been observed to proliferate in the injured retina (asterisks *SI Appendix, Fig. S1 A, Middle and Right hand*). Thus, by only counting proliferating cells in the INL, we can largely avoid these non-MG cell types. Importantly, pazopanib had no effect on cell death in the injured retina (*SI Appendix, Fig. S1 B*).

In the retina, MG are a source of Vegf ligands. Interrogation of previously generated retinal MG RNAseq data sets (13, 14) and RT-PCR assays revealed *vegfaa* and *pgfα* are the most highly expressed Vegf ligand-encoding genes in the injured retina, with *pgfα* exhibiting injury-dependent induction (Fig. 1B and *SI Appendix, Fig. S1 C and D*).

## Significance

Degeneration of retinal neurons underlies a variety of blinding eye diseases. Thus, restoring lost neurons is a major goal of vision scientists. One approach is to use endogenous stem cells to regenerate damaged retinal neurons. Unlike mammals, zebrafish can regenerate a damaged retina. Key to this regenerative response are Müller glia (MG), a retinal cell type found in both fish and mammals. Understanding how MG attain stem cell properties in zebrafish may suggest strategies for stimulating a similar regenerative response by human MG. Here, we report that Vegf signaling is critical for imparting zebrafish MG with stem cell properties and that this signaling pathway links immune cells and vascular endothelial (VE) cells with MG reprogramming and proliferation.

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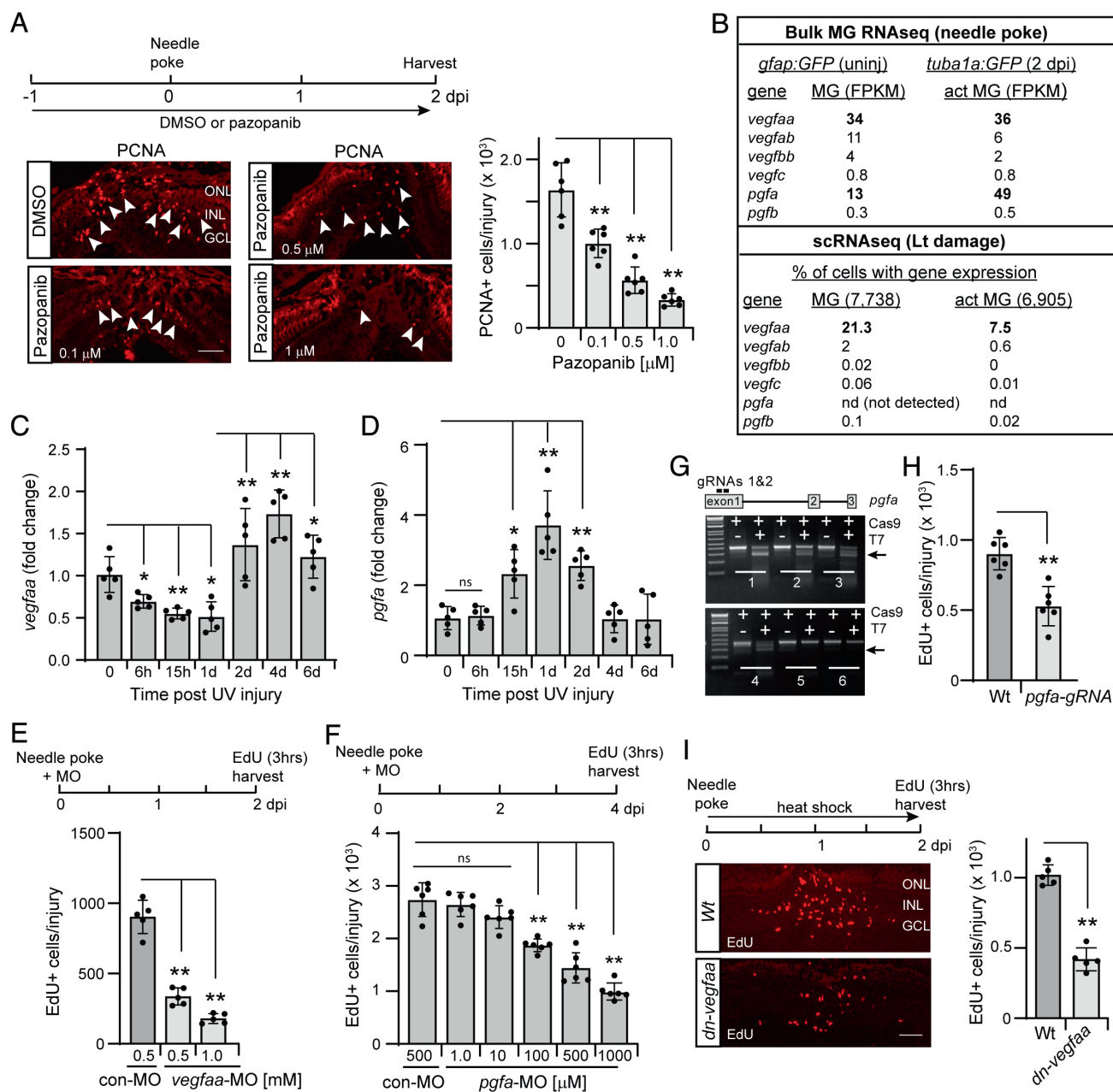
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RT-qPCR using RNA from retinas damaged by UV light revealed a transient suppression of *vegfaa* expression that preceded its induction around 2 dpi, while *pgfa* expression began to increase between 6 h post injury (hpi) and 15 hpi (Fig. 1 C and D).

To examine if other retinal cell types are also a source of Vegf ligands, we interrogated previously generated retinal scRNAseq data sets covering 0 to 36 hpi (2). This analysis revealed *vegfaa* is enriched in the MG population but suppressed in activated MG

around 20 to 36 hpi (Fig. 1B and SI Appendix, Fig. S1C). Surprisingly, *pgfa* expression was not detected in the scRNAseq data. Nonetheless, our bulk MG RNAseq (Fig. 1B), RT-PCR assays (SI Appendix, Figs. S1D and S2A), (BAC)*vegfaa:EGFP* transgenic fish (SI Appendix, Fig. S2B), and in situ hybridization assays (SI Appendix, Fig. S2D), all indicate *pgfa* and *vegfaa* are injury responsive genes that may be responsible for regulating Vegf signaling and MG proliferation.



**Fig. 1.** Vegfaa and Pgfa regulate MG proliferation in the injured retina. (A), The pan Vegf inhibitor, pazopanib, inhibits injury-dependent MG proliferation. Top is experimental timeline. Below timeline is representative images showing retinal PCNA immunofluorescence in fish treated +/- pazopanib (arrow heads point to PCNA+ cells in INL). Graph is quantification of PCNA+ immunofluorescence in fish treated with various concentrations of pazopanib. (B), Summary of Vegf ligand gene expression from previously published retinal MG RNAseq and scRNAseq data sets (2, 14). (C, D), RT-qPCR of *vegfaa* (C) and *pgfa* (D) RNA expression in uninjured and injured retina. (E, F), EdU Click-iT assay for proliferating cells in control and either Vegfaa knockdown (E) or Pgfa knockdown (F) retina. (G), Top is the diagram of *pgfa* gene with gRNA target sites indicated (small black rectangles above exon 1). Below is PCR using tail fin genomic DNA from adult CRISPR-based gene-edited fish in T7E1 assay to identify F0 fish with *pgfa* gene edits. Fish numbers 1 to 4 exhibited good gene editing and were used to assay the effects of retinal injury on MG proliferation. Arrow points to DNA with gene edits. (H), Quantification of EdU+ cells in Wt and CRISPR-based gene edited *pgfa* mutant fish. (I), Top is experimental timeline. Below the timeline are representative images showing EdU+ cells on heat shock-treated wild-type and *hsp70:dn-vegfaa* transgenic fish. Graph is quantification of EdU+ cells in heat shock-treated Wt and *hsp70:dn-vegfaa* transgenic fish retinas. Error bars are SD. Significant P values are \*P < 0.05, \*\*P < 0.01. The number of biological replicates is indicated by dots in each graph. Scale bars are 80 microns. Abbreviations: ns, nonsignificant; ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer; Con, control; MO, morpholino; dn, dominant-negative; Wt, wild type.

To investigate if Vegfaa and Pgfα regulate MG proliferation in the injured retina, we knocked down their expression with either a previously validated translation blocking *vegfaa* targeting MO (*vegfaa*-MO) (29, 30), or a translation blocking *pgfα*-MO. MO effectiveness was confirmed by injecting single-cell zebrafish embryos with chimeric *vegfaa*-GFP or *pgfα*-GFP RNA along with control (con)-MO or experimental MO. Reduced GFP fluorescence in experimental MO-treated embryos indicated effective knockdown (*SI Appendix*, Fig. S1 E and F). In the adult retina, Vegfaa or Pgfα knockdown reduced injury-dependent MG proliferation (Fig. 1 E and F and *SI Appendix*, Fig. S1 G). In addition, CRISPR-mediated *pgfα* gene editing in F0 fish also revealed reduced MG proliferation at 2 dpi (Fig. 1 G and H). Furthermore, heat shock treatment of *hsp70:dn-vegfaa* transgenic fish expressing a mutant Vegfaa protein that blocks the action of all Vegf ligands (31) suppressed injury-dependent MG proliferation (Fig. 1I). Importantly, these effects were not accompanied by changes in retinal cell death (*SI Appendix*, Fig. S1 H–J).

### Vegfrs on VE Cells Regulate MG Proliferation in the Injured Retina.

To identify cells responding to Vegf ligands, we interrogated MG RNAseq and retinal scRNAseq data sets for Vegf expression. This analysis revealed MG express *kdr* and *kdr1*, while VE cells predominantly express *flt1* and *kdr1*, with *flt1* expression uniquely restricted to VE cells (Fig. 2 A and B). Supporting this, we found *kdr1* and *flt1* highly enriched in FACS-purified GFP+ VE cell population isolated from *flt1:EGFP* transgenic fish retina (*SI Appendix*, Fig. S2A). In addition, *-0.8flt1:RFP* transgenic fish revealed RFP expression selectively in VE cells (*SI Appendix*, Fig. S2C). Furthermore, RNAseq and RT-PCR data indicate *kdr* is an injury-responsive gene (Fig. 2A and *SI Appendix*, Fig. S3 A and B). Because most *kdr1* expression and essentially all *flt1* expression is confined to VE cells, and because VE cells comprise a very small fraction of retinal cell types, these RNAs were poorly amplified following RT-PCR with total retinal RNA (*SI Appendix*, Fig. S3A).

Injury-dependent induction of *kdr* suggests it may mediate the effects of Vegf ligands on MG proliferation. Surprisingly, *Kdr* knockdown with a previously validated splice blocking MO had no significant effect on injury-dependent MG proliferation (*SI Appendix*, Fig. S3C) (32). We confirmed *Kdr* knockdown in the adult retina by RT-PCR using total retinal RNA (*SI Appendix*, Fig. S3D). In contrast, *Kdr1* knockdown with a previously validated translation blocking MO resulted in reduced injury-dependent MG proliferation without affecting cell death (*SI Appendix*, Fig. S3 E and F) (33). However, because *kdr1* is detected in both MG and VE cells (Fig. 2 A and B), we could not differentiate if one or both cell types contribute to Vegf-mediated MG proliferation in the injured retina.

Unlike *kdr1*, *flt1* expression is restricted to VE cells and provides an opportunity to test if Vegf signaling through VE cells regulates retina regeneration. For this analysis, we took advantage of a translation blocking *flt1*-MO to knockdown Flt1 expression. *flt1*-MO effectiveness was tested by injecting single-cell zebrafish embryos with a plasmid expressing a chimeric *flt1*-GFP RNA along with con-MO or *flt1*-MO. Reduced GFP fluorescence in *flt1*-MO-treated embryos indicated effective knockdown (*SI Appendix*, Fig. S3G). Remarkably, Flt1 knockdown in the adult retina reduced injury-dependent (needle poke) MG proliferation without affecting cell death (Fig. 2C and *SI Appendix*, Fig. S3H). These results were recapitulated in *flt1<sup>bms29</sup>* mutant zebrafish that harbor a premature stop codon at tyrosine 86 (Fig. 2D and *SI Appendix*, Fig. S3I) (34). The *flt1<sup>bms29</sup>* mutation also suppressed MG proliferation in N-methyl-D-aspartate (NMDA)-damaged and ultraviolet (UV) light-damaged retinas (*SI Appendix*, Fig. S3 J and K).

**Vegf Signaling via Flt1 Receptors on VE Cells Regulates Retinal Neuron Regeneration.** We next investigated if the reduced proliferation in *flt1<sup>bms29</sup>* fish impacted regeneration of retinal neurons. For this, we used an EdU-based lineage tracing strategy. Wt or *flt1<sup>bms29</sup>* fish retinas were injured, and proliferating MG were labeled with EdU at 4 dpi and then sacrificed 2 wk post injury when most MG progenitors have ceased proliferating (5). Retinal sections were stained and quantified for EdU and retinal neuron type-specific markers (*SI Appendix*, Fig. S4 A–C). This analysis revealed reduced MG proliferation and neuron regeneration in *flt1<sup>bms29</sup>* fish (*SI Appendix*, Fig. S4A). However, the relative proportion of different neurons regenerated from EdU+ progenitors in *flt1<sup>bms29</sup>* fish remained like Wt fish (*SI Appendix*, Fig. S4B). Thus, Vegfaa and Pgfα appear to act via Flt1 receptors on VE cells to regulate MG proliferation and neuron regeneration.

### Vegf Signaling through VE Cells Regulates Notch Signaling in MG.

Notch signaling stimulates MG quiescence and raises MG's injury response threshold (7, 11, 13, 14, 17, 35). We previously reported that *dll4* and *hey1* expression levels reflect Notch signaling in MG and regulate MG proliferation (11, 14). Furthermore, *dll4* expression is enriched in VE cells, while Notch3 and *hey1*, a direct Notch target gene, are enriched in MG (Fig. 3A) (14, 17). scRNAseq data indicate injury-dependent *dll4* repression in VE cells (14), and we verified this using *flt1:EGFP* transgenic fish that allowed us to purify GFP+ VE cells by FACS (*SI Appendix*, Fig. S4D) (36). RT-PCR for *kdr1* and *ascl1a*, who are highly enriched in VE cells and MG, respectively, confirmed successful FACS-based VE cell enrichment (*SI Appendix*, Fig. S4E).

We next tested if Vegf signaling impacted *dll4* in VE cells and *hey1* in MG using RT-qPCR and total retinal RNA from uninjured retinas. Inhibition of Vegf signaling with pazopanib, forced expression of *dn-vegfaa*, or with *flt1<sup>bms29</sup>* mutant fish, resulted in increased *dll4* and *hey1* gene expression (Fig. 3 B–D). Importantly, FACS purification of MG from *gfap:GFP* fish confirmed increased *hey1* expression in MG following Vegf inhibition with pazopanib (Fig. 3E). Together, these data suggest Vegf signaling in VE cells suppresses *dll4* to reduce Notch signaling and *hey1* expression in MG.

### Vegf Signaling Regulates a Portion of the MG Regeneration-Associated Transcriptome.

To get a better idea of the impact Vegf signaling has on regeneration-associated genes in the injured retina, we compared injury-dependent regulation of select genes in Wt and *flt1<sup>bms29</sup>* mutant fish. Surprisingly, RT-qPCR showed injury-dependent induction of *ascl1a*, *lin28a*, and *myh* remained largely unaffected by the *flt1<sup>bms29</sup>* mutation (Fig. 3F), while induction of *hbegfa*, *ccna2*, and *pcna* was abrogated (Fig. 3G). Further, the normal injury-dependent suppression of *hey1*, *fgf8a*, and *col15a1b* was not observed in the *flt1<sup>bms29</sup>* mutant retina (Fig. 3H).

We were surprised that reprogramming genes, like *ascl1a* and *lin28a* remained induced in the injured *flt1<sup>bms29</sup>* mutant fish retina (Fig. 3F). To further investigate this, we took advantage of *6-ascl1a:GFP* transgenic fish that harbor a 6-kb fragment of the *ascl1a* promoter and retains injury-dependent regulation (9). Like Wt fish, injury-dependent GFP expression was not suppressed in pazopanib-treated *6-ascl1a:GFP* fish, while *ccna2* and *pcna* expression was reduced by this treatment (Fig. 3I). Together, these data suggest Vegf signaling via Flt1 receptors on VE cells regulates Notch signaling and the expression of a subset of regeneration-associated genes in MG.

### Microglia/Macrophages Regulate Vegf Ligand Expression in MG.

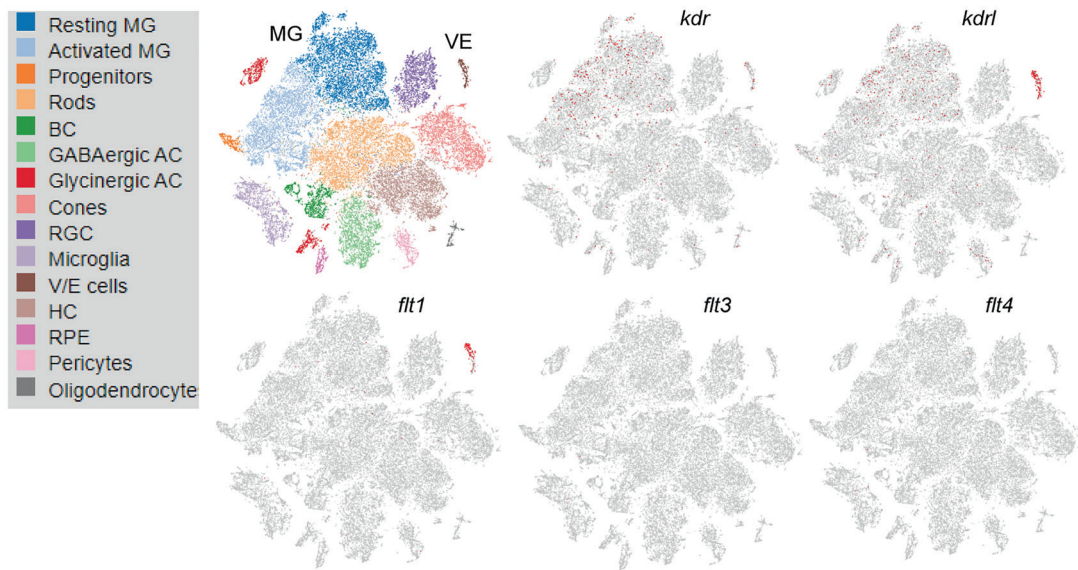
Following retinal injury, microglia/macrophages accumulate at the injury site and stimulate MG proliferation (37–39). Colony-stimulating factor 1 receptor (Csf1r) is necessary for maintaining



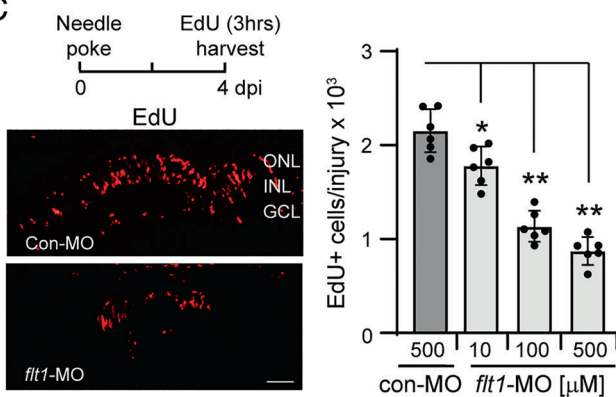
A

| Bulk MG RNAseq (needle poke) <sup>14</sup> |           |                          | scRNAseq (Lt damage) <sup>2</sup> |                |                 |
|--------------------------------------------|-----------|--------------------------|-----------------------------------|----------------|-----------------|
| <i>gfap:GFP</i> (uninj)                    |           | <i>tuba1a:GFP</i> (2dpi) | % of cells with gene expression   |                |                 |
| gene                                       | MG (FPKM) | act MG (FPKM)            | MG (7,738)                        | act MG (6,905) | V/E cells (427) |
| <i>kdr</i>                                 | 11.1      | 22.7                     | 2.0                               | 1.6            | 3.5             |
| <i>kdr1</i>                                | 1.2       | 1.2                      | 1.9                               | 1.0            | 37.7            |
| <i>flt1</i>                                | 0.1       | 0.1                      | 0.13                              | 0.03           | 42.9            |
| <i>flt3</i>                                | 0.09      | 0.05                     | 0.001                             | 0.01           | 0               |
| <i>flt4</i>                                | 0.03      | 0.22                     | 0.02                              | 0              | 5.4             |

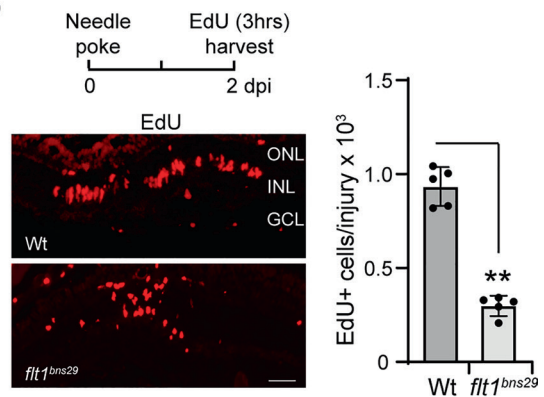
B



C



D

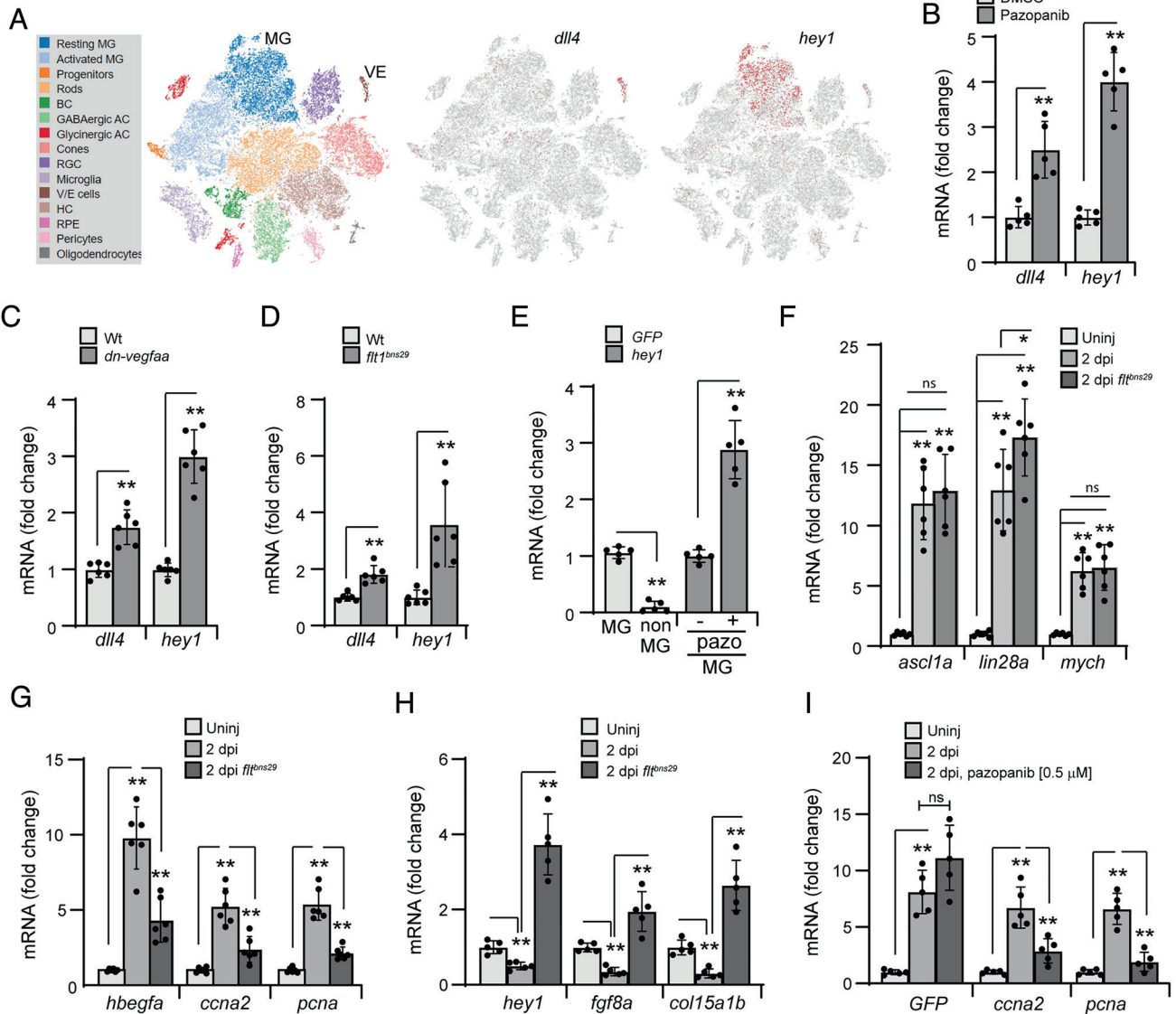


**Fig. 2.** Flt1 receptors regulate MG proliferation in the injured retina. (A), Summary of Vegf receptor gene expression from previously published retinal scRNAseq and MG RNAseq data sets (2, 14). (B), UMAP plot showing various retinal cell types in the UV light-damaged retina with gene expression of various Vegf receptors overlaid (red). scRNAseq data are from Hoang et al., 2020. (C), Top is experimental timeline. Below are representative images of retinal sections showing EdU+ cells in fish treated with con-MO or *flt1*-MO. Graph is quantification of EdU+ cells/injury after treatment with indicated concentrations of *flt1*-MO. (D), Top is experimental timeline. Below are representative images of retinal sections showing EdU+ cells in Wt and *flt1*<sup>bns29</sup> mutant zebrafish. Graph is quantification of EdU+ cells/injury. Significant *P* values are \**P* < 0.05, \*\**P* < 0.01. Scale bars are 80 microns. Error bars are SD. The number of biological replicates is indicated by dots in each graph. Abbreviations as in Fig. 1.

the CNS microglia/macrophage population (40, 41), and these cells can be detected with the 4C4 antibody (42). *csf1r*<sup>DM</sup> fish and Wt fish treated with the Csf1r inhibitor, PLX3397, exhibit a significant reduction in 4C4+ cells at the injury site that is paralleled by a suppression in MG's proliferative response to retinal injury (Fig. 4 A–F). Similar results were obtained when fish were immersed in the pancaspase inhibitor, ZVAD-fmk, which inhibits apoptosis and suppresses microglia/macrophage accumulation (Fig. 4 G–J). Importantly, fish with compromised Vegf signaling, like *flt1*<sup>bns29</sup> and heat shock-treated *hsp70:dn-vegfaa* fish, exhibit

reduced injury-dependent MG proliferation (Figs. 1I and 2D) without affecting microglia/macrophage (4C4+) accumulation at the injury site (Fig. 4 J–L).

We next investigated if microglia/macrophage accumulation at the injury site engaged the Vegf signaling pathway to regulate injury-dependent MG proliferation. Indeed, reducing microglia/macrophage recruitment to the injury site using either *csf1r*<sup>DM</sup> mutant fish or Wt fish immersed in ZVAD-fmk, revealed reduced injury-dependent induction of *vegfaa* and *pgfa* gene expression (Fig. 5 A and B). It was previously reported that these immune



**Fig. 3.** Vegf signaling regulates Notch signaling and a portion of the regeneration-associated transcriptome in MG. (A), UMAP plots showing various retinal cell types in the light-damaged retina with *dll4* and *hey1* gene expression overlaid (red). scRNAseq data are from Hoang et al., 2020. (B–D), Total retinal RNA from uninjured retina was used for RT-qPCR quantification of *dll4* and *hey1* gene expression in Wt fish treated with DMSO or pazopanib (B), Wt and *hsp70:dn-vegfaa* transgenic fish treated with heat shock (C), and Wt and *flt1<sup>bns29</sup>* mutant fish (D). (E), GFP+ MG and GFP- nonMG were FACS purified from pazopanib (500 nM)-treated or DMSO-treated *gfap:GFP* fish. RT-qPCR was used to assay *GFP* expression in GFP+ MG and GFP- nonMG populations to ensure successful cell sorting. *hey1* gene expression was assayed in the GFP+ MG population. (F–H), RT-qPCR quantification of the expression of select regeneration-associated genes in uninjured and injured Wt retina, and injured *flt1<sup>bns29</sup>* mutant retina. (I), RT-qPCR quantification of *GFP*, *ccna2*, and *pcna* gene expression using total RNA from uninjured and injured (2 dpi) DMSO or pazopanib-treated *6-ascl1a:GFP* fish retina. Error bars are SD. Significant *P* values are \**P* < 0.05, \*\**P* < 0.01. The number of biological replicates is indicated by dots in each graph. Abbreviations as in Fig. 1.

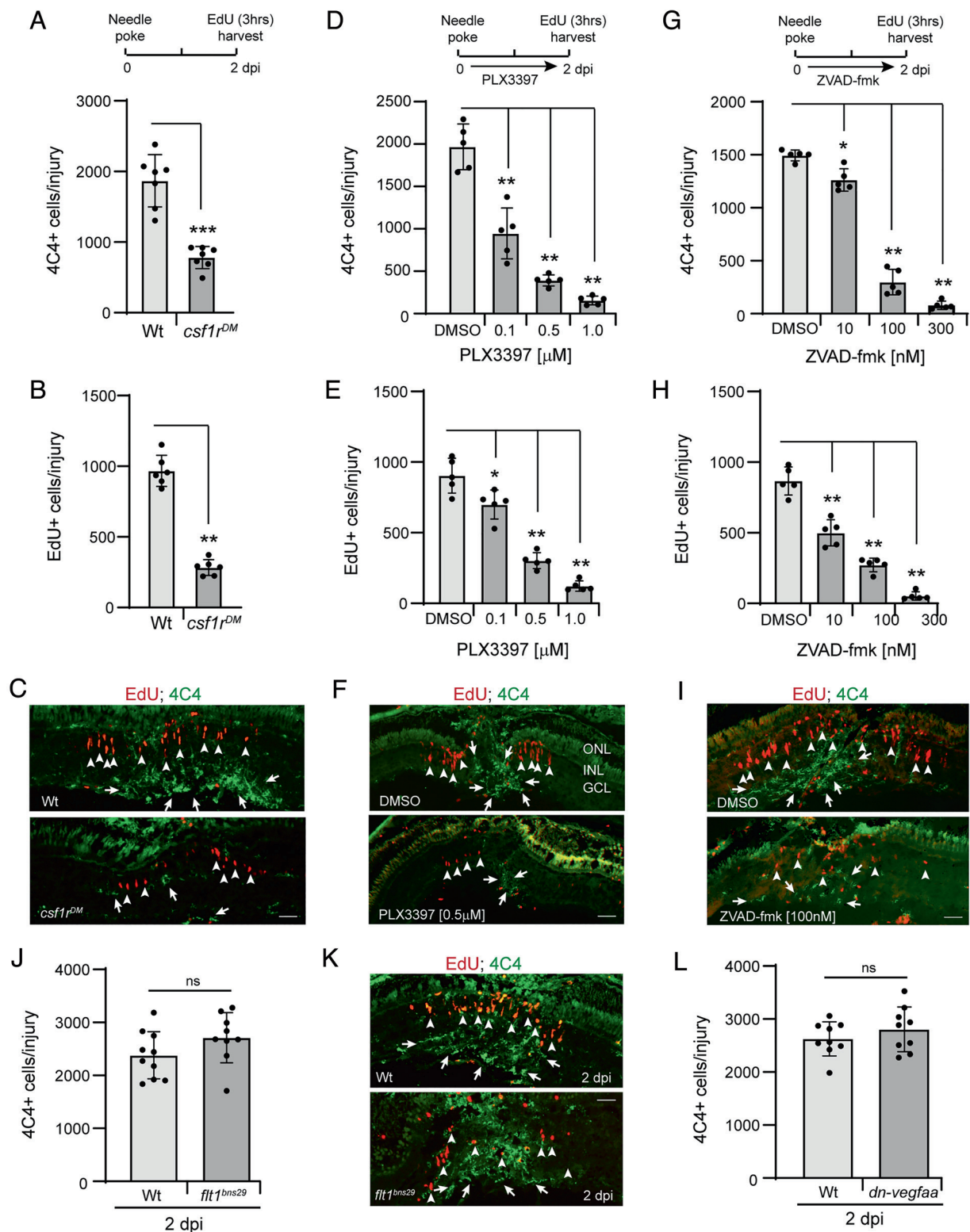
cells stimulate mTOR signaling in MG (20), and we found that mTor inhibitor, rapamycin also suppresses injury-dependent induction of *vegfaa* and *pgfa* (Fig. 5C). Finally, like inhibition of Vegf signaling, we found microglia/macrophage ablation in *csflr<sup>DM</sup>* fish prevented or reversed injury-dependent changes in *dll4*, *hey1*, *hbegfa*, *mych*, *pcna*, and *ccna2* gene expression, without affecting *ascl1a* and *lin28a* induction (Fig. 5D–F).

## Discussion

The above studies identify a Vegf signaling system that regulates MG proliferation in the injured zebrafish retina. Importantly, this signaling system is regulated by immune cells and links MG to VE cells (Fig. 5G). Remarkably, this signaling system not only regulates *dll4* expression in VE cells but also regulates a subset of

regeneration-associated genes in MG, including those involved in Notch signaling and cell proliferation.

Vegf signaling has been best studied in the context of the vasculature where it regulates angiogenesis via its actions on endothelial cells 43–45. During development, Vegf engages Kdr on VE tip cells to enhance *Dll4* expression, Notch signaling, and angiogenic sprouting, while engagement of Flt1 on these cells suppresses Notch signaling and angiogenic sprouting 45, 46. Our studies indicate MG-derived Vegfaa and Pgfa engage Flt1/Kdr on mature VE cells to inhibit *dll4* expression in VE cells and Notch signaling in MG. *Dll4* likely mediates its effects on MG via Notch3 receptors expressed by MG and necessary for regulating MG quiescence (14, 17). Together these data suggest that Vegf ligand-receptor pairing, and VE cell maturation influences downstream signaling that dictates the biological response.

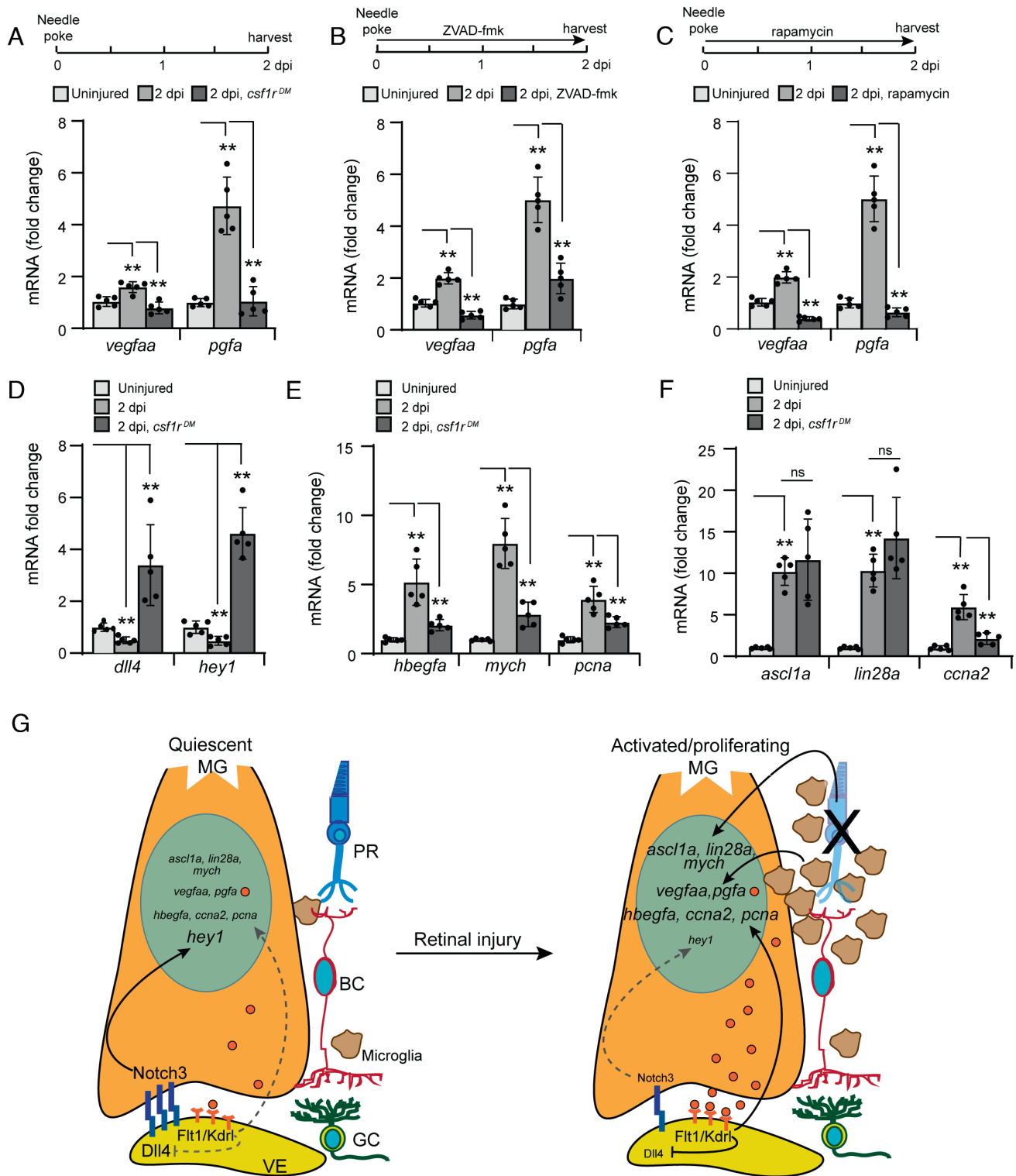


**Fig. 4.** Vegf signaling does not regulate microglia/macrophage accumulation in the injured retina. (A, D, G), Quantification of 4C4 immunofluorescence in injured Wt and *csf1r<sup>DM</sup>* retinas (A), PLX3397-treated retina (D), and ZVAD-fmk-treated retina (G). (B, E, H), Quantification of EdU+ cells in injured Wt and *csf1r<sup>DM</sup>* retinas (B), PLX3397-treated retina (E), and ZVAD-fmk-treated retina (H). (C, F, I), Representative fluorescence images of retinal sections showing effects of *csf1r<sup>DM</sup>* mutation (C), PLX3397 (F), and ZVAD-fmk (I) on EdU (red) and 4C4 (green) immunofluorescence. Arrows and arrowheads point to 4C4+ and EdU+ cells, respectively. (J), *flt1<sup>bns29</sup>* mutant zebrafish exhibit normal microglia/macrophage (4C4+) accumulation at the injury site. (K), Representative fluorescence image of EdU+ and 4C4+ cells at the injury site of Wt and *flt1<sup>bns29</sup>* mutant zebrafish. Arrows and arrowheads point to 4C4+ and EdU+ cells, respectively. (L), Forced expression of *dn-vegfaa* in *hsp70:dn-vegfaa* transgenic fish has no effect on microglia/macrophage (4C4+) accumulation at the injury site. Error bars are SD. Significant *P* values are \**P* < 0.05, \*\**P* < 0.01, \*\*\**P* < 0.001. Size markers are 80 microns. The number of biological replicates is indicated by dots in each graph. Abbreviations are as in Fig. 1.

Importantly, Vegf signaling through Flt1 receptors on VE cells not only impacts Notch signaling and gene expression in MG but may also influence local vasculature permeability influencing

the availability of blood-derived factors that affect MG behavior. The transient activation of Vegf signaling following retina injury may be critical for preventing pathology. In the adult retina,





**Fig. 5.** Microglia/macrophages regulate *vegfaa* and *pgfa* expression, along with a portion of the regeneration-associated transcriptome, in MG. (A–C), Top diagram is experimental timeline. Below is RT-qPCR quantification of *vegfaa* and *pgfa* gene expression in uninjured and injured Wt, and injured *csf1r<sup>DM</sup>* fish retinas (A); uninjured and injured Wt fish +/- ZVAD-fmk (100  $\mu$ M) (B), and uninjured and injured Wt fish +/- rapamycin (25  $\mu$ M) (C). (D–F), RT-qPCR quantification of select regeneration-associated genes in uninjured and injured Wt and *csf1r<sup>DM</sup>* mutant fish retinas. (G), Illustration summarizing data presented in this report. Notch signaling is most active in quiescent MG, while Vegf signaling is most active in activated/proliferating MG (solid lines with arrows). Font size indicates gene expression level. Solid arrows indicate signaling activation, while dashed arrows indicate reduced signaling. X indicates neuron injury. A role for Notch3 in mediating MG Notch signaling is indicated in Sahu et al., 2021 and Campbell et al., 2021 (14, 17). Abbreviations: MG, Muller glia; VE, vascular endothelial cell; PR, photoreceptor; BC, bipolar cell; GC, ganglion cell. Error bars are SD. Significant *P* values are  $^{***}P < 0.01$ . The number of biological replicates is indicated by dots in each graph. Abbreviations are as in Fig. 1.

dysregulation of Vegf signaling contributes to vascular overgrowth and changes in vascular permeability that are associated with macular degeneration and diabetic retinopathy 47–49.

Whether blood-derived factors are involved in mediating the effects of Vegf signaling on retina regeneration remains to be determined.

Our studies suggest that MG-derived Vegfaa and Pgfα engage Flt1/Kdr receptors on VE cells to regulate MG reprogramming and proliferation. Although Flt1 receptors bind these ligands with high affinity, they generally exhibit weak phosphorylation suggesting they may function as a Vegf ligand sink 45. However, signaling through Flt1 receptors does occur in certain circumstances and may be developmental stage, cell type, and/or ligand type dependent 45. Furthermore, *flt1* is coexpressed with *kdr1* in VE cells allowing for a heterodimeric high-affinity receptor with strong phosphorylation and signaling. Consistent with this idea, Kdr1 and Flt1 knockdown have similar effects on MG proliferation in the injured retina.

Interestingly, *Vegfa*, *Vegfb*, and *Pgfα* are expressed in the mouse retina (2). However, unlike *vegfaa* and *pgfa* in the fish retina whose expression is enriched in MG, mouse *Vegfa* and *Pgfα* are expressed more uniformly in multiple cell types (SI Appendix, Fig. S5). Furthermore, unlike the zebrafish retina where *flt1* is restricted to VE cells, in the mouse retina *Flt1* is more highly expressed in MG, raising the possibility that it may serve as a Vegf ligand sink and thereby impacts Vegf signaling in VE cells. Whether this contributes to the poor regenerative response of mammalian MG remains to be investigated.

We previously reported that *dll4* in VE cells is regulated by retinal injury and proposed it may contribute to Notch signaling in MG (14). We also reported that MG expression of *hey1* is directly regulated by Notch signaling (14). Here we extend these findings by showing *dll4* in VE cells, and *hey1* expression in MG is regulated by Flt1 receptors on VE cells. Thus, basal Vegf signaling in VE cells regulates Notch signaling in MG and thereby contributes to MG quiescence. Furthermore, enhanced Vegf signaling after retinal injury not only reduces Notch signaling in MG, but also increases the expression of genes, like *hbegfa*, *ccna2*, and *pcna* that stimulate MG proliferation. Importantly, our studies reveal that microglia/macrophage and VE cells only regulate a portion of the regeneration-associated transcriptome. Injury-regulated genes, like *ascl1a* and *lin28a* were not regulated by Vegf signaling or microglia/macrophage ablation. The fact that strong reprogramming genes like *ascl1a* and *lin28a* are induced in MG independent of microglia/macrophage accumulation suggests that their induction may be controlled by dying neurons. Although the significance of this partial reprogramming remains unknown, it may prime MG so they can enter a regenerative response upon microglia/macrophage accumulation. Indeed, Ascl1 has been shown to enhance chromatin accessibility, and Lin28 contributes to the induction of pluripotency 50,51. Furthermore, we previously reported that forced expression of *Ascl1a* and *Lin28a* stimulates a small number of MG to proliferate in the uninjured retina and that this is greatly enhanced following retinal injury (12). Thus, the maintained expression of these genes when Vegf signaling is compromised may contribute to the relatively small amount of proliferation noted under these conditions. However, our previous overexpression studies achieved *ascl1a* and *lin28a* levels that exceeded those normally found in the injured retina, and it remains to be determined if physiological levels of *Ascl1a* and *Lin28a* are sufficient to drive some MG proliferation. Regardless, this 2-step reprogramming process ensures MG only fully reprogram and enter a highly proliferative state when dying neurons and cellular debris can be cleared by accumulating phagocytic cells at the injury site.

Unlike the injured mammalian retina where microglia/macrophages inhibit MG transdifferentiation without affecting their proliferation 52, these cells are a critical component of the regenerative niche that stimulates mTOR signaling and MG proliferation in the injured zebrafish retina (20, 37). Our studies extend these

findings and suggest that mTOR acts, at least in part, via a VE cell-dependent Vegf signaling pathway. The mechanism by which microglia/macrophage engage mTOR signaling in zebrafish MG and whether these cells engage a Vegf signaling pathway in the mammalian retina remains unknown.

We were surprised that Kdr receptor knockdown had no effect on MG's proliferative response following retinal injury. This suggests another role for Kdr that may impact MG homeostasis, survival, migration, and/or differentiation. Indeed, Vegf has been shown to regulate MG and neuron survival in the adult mouse retina 53. Further studies are needed to address Kdr's function in MG.

Finally, we note reports suggesting increased Vegf signaling can stimulate neural progenitor cell proliferation in the postnatal degenerating rd1 mouse retina and in the adult CNS 54–57. Thus, this pathway along with other proregenerative factors identified to control retina regeneration in zebrafish may help facilitate a similar process in mammals.

## Materials and Methods

**Experimental Models.** All animal work was approved by the University of Michigan's Institutional Animal Care and Use Committee. Zebrafish were housed in a recirculating water system maintained at ~26 °C. Room lights were on a 14/10-h light/dark cycle. This study used male and female fish at 6 to 12 mo of age. Transgenic and mutant fish lines used in this study are: *hsp70:dn-vegfaa121-F17A<sup>bns100</sup>*, *fli1:EGFP*, *flt1<sup>bns29</sup>*, *csf1ra<sup>-/-</sup>/csf1rb<sup>-/-</sup>* (called *csf1r<sup>DM</sup>*), (*BAC*)*vegfaa:EGFP*, *gfap:GFP*, *-0.8flt1:EGFP*, and *6-ascl1a:GFP* 5,9,34,58–64. All lines, except for the *6-ascl1a:GFP* fish, were on the AB background. *6-ascl1a:GFP* fish were generated using a pet store-derived Wt fish line that was inbred for over 30 generations. Control Wt fish used for comparison with transgenic and mutant lines are on the AB genetic background. Fin clips were used to isolate genomic DNA for genotyping fish. Briefly, fin clips were collected in 50 μL lysis buffer (10 mM Tris-HCl pH8.0, 2 mM EDTA, 0.2% Triton X-100 and 100 mg/mL Proteinase K) and incubated at 55 °C for 1 h with frequent vortexing and trituration. Samples were then heated to 95 °C for 10 min and insoluble material removed by centrifugation. Then, 0.5 μL genomic DNA was amplified by RT-PCR using gene-specific primers (SI Appendix, Table S1), and GoTaq Green Master Mix (Promega M7122) or Radiant 2× RED Taq mix (Alkali Scientific C225) in a 25-μL reaction volume. PCR products (7 μL) were analyzed on agarose gels with DNA size standards for determining amplified DNA size. For detecting gene edits, very small deletions, or fish harboring heterozygous mutations, a 7-μL PCR sample was used in T7E1 assay (+/-T7) as previously described 65.

Fish anesthesia was carried out by immersing fish in tricaine containing fish water (75 mg/L) buffered to pH7.2. Retinal injuries were either needle poke (1 to 4 injuries/retina for proliferation and immunofluorescence assays on retinal sections, and 8 injuries/retina for whole retina RNA isolation), intense UV light, or intravitreal NMDA injection as previously described 5, 14, 66, 67.

**Cell Proliferation Assays.** PCNA immunofluorescence was performed with mouse anti-PCNA antibody (Sigma, 1/500 dilution) on 8-micron retinal sections as previously described 68. For EdU labeling of proliferating cells, fish were injected intraperitoneally (IP) with 10 μL EdU (10mg/mL) 3 h before sacrifice. Click-It chemistry was used to detect EdU in retinal sections (Molecular Probes, 1511352). PCNA+ and EdU+ cells were quantified in the INL where MG and proliferating MG-derived progenitors reside (5).

**RNA Isolation and PCR.** Fish were dark adapted overnight, anesthetized in tricaine, and then eyes were harvested and placed in a tray of ice-cold phosphate-buffered saline (PBS) for retinal dissection. Dissected retinas (generally 3 replicates) were placed into 400 μL TRIzol (Invitrogen) and dissociated by trituration using, first a P200 Pipettman, and then a 1-mL syringe equipped with a 30-g needle. RNA was then purified using the manufacturer's (TRIzol, Invitrogen) recommendations. RNA concentration was determined on a NanoDrop One spectrophotometer (Thermo Fisher), and ~1 μg RNA was used for cDNA synthesis using M-MLV or SSIII reverse transcriptase (Invitrogen) according to the manufacturer's directions. cDNA was diluted 1/5 and 1 μL was used for PCR as previously described 69,70. We used a Luna qPCR Master Mix (NEB) and an iCycler real-time



PCR detection system (BioRad) to carry out real-time PCR. The  $\Delta\Delta C_t$  method was used to determine mRNA expression levels, and this was normalized to *gapdh<sub>s</sub>* for determining fold change in RNA expression.

**CRISPR-Based Gene Editing.** We used CRISPRscan to identify 6 gRNAs that target *pgfa* genomic DNA 71. gRNA DNA template synthesis using gRNA primer, universal primer, and in vitro transcription has been described 72. gRNA sequences are listed in [SI Appendix, Table S1](#). To identify the best gRNAs for gene editing, we injected single-cell zebrafish embryos with individual in vitro transcribed gRNAs and Cas9-nanos encoding RNA. Two days later, some embryos were used to determine gene-editing efficiency using T7 endonuclease 1 (T7E1) mismatch detection assays 65. The remaining embryos from groups showing high gene-editing efficiency were raised to adults to confirm efficient gene editing using genomic DNA isolated from fin clips in T7E1 mismatch assays 65. CRISPR/Cas9 gene-edited adult fish with ~50% or more gene editing were used in these studies.

**PCR Primers, Morpholino (MO), and gRNAs.** See [SI Appendix, Table S1](#) for PCR primers, MOs, and gRNAs used in this study.

**MO Functional Assays.** Lissamine-tagged MOs were obtained from Gene Tools, LLC. Splice blocking *kdr*-MO was validated by injecting control MO or *kdr*-MO (~1 ng) into the vitreous of adult eyes followed by electroporation to facilitate MO uptake by cells as previously described 69. RNA was extracted 2 d later and assayed for *kdr* mRNA expression by RT-PCR and agarose gel electrophoresis. Translation blocking MOs were validated by generating *pTAL-sCMV-MO target site-EGFP* chimera that have the MO target site cloned upstream of the AUG that initiates EGFP translation. For this, we amplified the region spanning the MO binding site using retinal RNA and BamH1-forward and Age1-reverse primers ([SI Appendix, Table S1](#) "Primers for chimeric RNA"). BamH1 and Age1 were then used to restrict the amplified DNA and the pTAL-sCMV-EGFP plasmid, and gel purified DNA was used in the ligation reaction. Successful cloning was confirmed by DNA sequencing. Kpn1 was used to linearize the vector and capped sense RNA was synthesized using Invitrogen's mMACHINE™ SP6 Transcription Kit (Invitrogen, #AM1340). Fifty picograms of purified capped RNA along with 250 pg of control or experimental MO was injected into single-cell embryos and GFP fluorescence assayed 1 to 2 d later.

For gene knockdown in adult fish, MOs were injected into the eye's vitreous and electroporation used to facilitate cellular uptake 69.

**Heat Shock and Pharmacological Inhibitors.** *hsp70* promoters were activated by immersing fish in a 37 °C water bath for 1 h, which was repeated every 6 h for the duration of the experiment. For treating fish with various pharmacological reagents, stock solutions were dissolved in DMSO. Vegfr inhibition was achieved with pazopanib (SelleckChem, 10 mM stock) by incubating fish in system water containing a maximum of 1  $\mu$ M pazopanib starting 1 d prior to retinal injury and replacing with fresh pazopanib containing fish water daily (control: 0.0001% DMSO); mTOR signaling inhibition with rapamycin (Sigma, 1 mM stock) was achieved by daily IP injections of 15  $\mu$ L of a 25  $\mu$ M stock in PBS beginning a couple of hours before retina injury (control: PBS containing 2.5% DMSO); Csf1r was inhibited with PLX3397 (SelleckChem, 100 mM stock) by incubating fish in system water containing a maximum of 1  $\mu$ M PLX3397 starting 10 d prior to retinal injury and replacing with fresh PLX3397 containing fish water daily (control: 0.00001% DMSO); and caspase activity was inhibited with ZVAD-fmk (R&D Systems, 20 mM stock) by incubating fish in system water containing a maximum of 300 nM ZVAD-fmk just before retinal injury and replacing with fresh ZVAD-fmk containing fish water daily (control: 0.000015% DMSO).

**Immunofluorescence and TUNEL Assay.** Eight-micron retinal sections were prepared using a Leica CM 3050 S cryostat. Immunofluorescence was performed

as previously described 4,5,70. Primary antibodies used in this study: Rabbit anti-GFP, Thermo Fisher, Cat. # A6455 (1/1,000); mouse anti-PCNA Sigma, Cat# P8825 (1/500 dilution); mouse anti-4C4 (1/200 dilutions; gift from J. Parent lab, UM); rabbit anti-HuCD, Abcam, Cat# ab210554 (1/500); mouse anti-glutamine synthetase (GS), Millipore Sigma, Cat# MAB302 (1/500); mouse anti-zpr1, ZIRC, Cat# ZDB-ATB-081002-43 (1/500). Secondary antibodies: Alexa Flour 555 Donkey anti Mouse-IgG (H+L), Thermo Fisher Cat. # A31570 (1/500); Alexa flour 555 Donkey anti Rabbit IgG (H+L), Thermo Fisher, Cat # A31572 (1/500); Alexa Flour 488 donkey anti mouse Thermo Fisher Cat. # A21202 (1/500); Alexa Flour 488 goat anti rabbit Thermo Fisher Cat. # A11008 (1/500).

An in situ Cell Death Fluorescein Kit (Millipore Sigma, Cat. # 11684795910) was used to detect apoptotic cells.

**Microscopy and Cell Quantification.** Images were captured using a Zeiss Axio Observer inverted epifluorescence microscope equipped with an AxioCam MRm camera, or a Leica DM2500 epifluorescence microscope equipped with a DFC7000T camera using Zeiss Zen 2.5 (blue edition) and Leica LAS X software, respectively. Images were acquired using 20 $\times$  dry objective. Single focal plane images were collected and imported into Adobe Photoshop for global adjustments of brightness and contrast, and manual quantification of cell numbers. Images were then imported into Adobe Illustrator for labeling and creating multipanel figures. PCNA, EdU, 4C4, and TUNEL labeling was used to identify and quantify proliferating cells, microglia/macrophage, and apoptotic cells, respectively, in retinal sections as previously described 5,7,8,11,70. For quantification, each injury site is captured in multiple sections through the retina. Quantification of injury responsive cells (PCNA+, Edu+, TUNEL+, 4C4+, etc.) is counted on all sections that span the injury site, and the sum of these values is recorded and reported on the Y-axis in the graphs. In some cases, a representative image of a single section near the center of the injury site is shown to help visualize the injury response.

**Statistical Analysis.** Unless otherwise indicated, sample size is 3 retinas and experiments were repeated 5 to 6 times. Statistical analyses were performed in GraphPad Prism. The nonparametric two-tailed Mann-Whitney *U* test was used for pairwise comparisons and the Kruskal-Wallis test with Dunn's multiple comparison post hoc test was used for multiple group comparisons. Statistical significance was set at *P* < 0.05. All measurements were taken from distinct samples. Error bars are SD.

**Data, Materials, and Software Availability.** All study data are included in the article and/or [SI Appendix](#). Some study data available (Material requests and correspondence: Dan Goldman (neuroman@umich.edu)).

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